Spectral-Spatial Evidential Learning Network for Open-Set Hyperspectral Image Classification

Fengcheng Ji, Wenzhi Zhao, *Member, IEEE*, Qiao Wang, William J. Emery, *Life Fellow, IEEE*, Rui Peng, Yuanbin Man, Guoqiang Wang, and Kun Jia

***Abstract*— Deep learning-based classification methods of hyperspectral images (HSIs) have made significant progress recently, catching the attention of academia and industry; however, the existing studies of classification of HSIs mainly focus on the closed-set environment with the assumption that ground classes are fixed and known, ignoring the complexity and diversity of ground objects in the real world. As a result, the unknown classes will be forced into known classes. To solve this problem, we propose a novel spectral-spatial evidential learning (SSEL) network that combines an improved generative adversarial network (GAN) and evidential theory for open-set classification of HSIs. First, a domain adaptation (DA) strategy is embedded into GAN to generate high-quality samples by reducing the distribution discrepancy between generated and real samples. Second, the discriminator is devised to extract spectral-spatial features and output multiclass evidence for closed-set classification and uncertainty estimation. A new classification function called evidence-based loss is designed for the discriminator to guide the evidence-collection process. Additionally, a novel adversarial objective function is defined, where the discriminator loss is devised to predict real samples belonging to the true class and generated samples belonging to "none of the classes. The generator loss is developed to generate samples consistent with the label category. Finally, the class and corresponding uncertainty can be calculated based on the collected evidence and the appropriate open-set classification of HSIs. Extensive experiments on three benchmark HSIs show that our proposed method achieves competitive performance on closed-set and open-set classification of HSIs compared with existing state-of-the-art methods.**

***Index Terms*— Classification of hyperspectral images (HSIs), evidential learning (EL), generative adversarial network (GAN), open-set.**

Manuscript received 6 November 2023; revised 30 November 2023; accepted 21 December 2023. Date of publication 3 January 2024; date of current version 17 January 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62201063 and in part by the National Nature Science Foundation of China Major Program under Grant 42192580 and Grant 42192584. *(Corresponding author: Wenzhi Zhao.)*

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Digital Object Identifier 10.1109/TGRS.2024.3349415

I. Introduction

RECENTLY, the rapid development of hyperspectral sensors has achieved significant breakthroughs in the remote sensing field, which can now capture hundreds of contiguous and narrow spectral bands simultaneously, with wavelengths spanning the visible to the infrared spectrum. This wealth of information can provide both spatial and high spectral information from surface materials [1], [2], [3], [4]. Considering these characteristics, hyperspectral images (HSIs) have been widely used for purposes such as environmental monitoring [5], [6], mineral exploration [7], and precision agriculture [8]. Among these applications, classification of HSIs has attracted considerable attention in the remote sensing community due to its advantages in both spatial and spectral domains.

Over the past years, numerous traditional classification methods of HSIs have been proposed to improve classification accuracy. In the early stage, classification approaches of HSIs were mainly based on spectral linear transformations, such as principal component analysis (PCA) [9], discriminant analysis [10], [11], independent component analysis (ICA) [12], and etc., [13], [14]. These methods directly used the abundant spectral information to classify diverse land covers. Compared with the above methods, spectral classifier-based methods usually have a stronger generalization and nonlinear expression capability when applied to the classification of HSIs. Algorithms falling into this type include k-nearest neighbor [15], [16], random forest [17], and support vector machine (SVM) [18], [19]. Because of the high intraclass spectral variability and low interclass spectral variability, spectral-based methods, however, tend to struggle to discriminate between the different objects and further respond to unexpected noise.

Instead of only using spectral information for classification of HSIs, an increasing number of spectral-spatial classification methods, such as structural filtering [20], [21], [22], and sparse representation [23], [24], [25], integrate spectral signatures and spatial contexts and have emerged over the past few years. These methods take advantage of the vital spatial information provided by adjacent pixels in addition to the spectral information [26], [27], [28]. For structural filtering methods, they mainly use the spatial information around the pixel to extract the features related to the object category. For example, Shen and Jia [20] developed a 3-D Gabor-wavelet-based approach for the classification of HSIs, in which a set

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of complex Gabor wavelets with different frequencies and orientations were designed to extract the spatial information in local areas by calculating variances in spatial and spectral domains. The basic principle of the sparse representation method is under the assumption that the spectral-spatial response can be represented by a sparse linear combination of a few basis vectors in the training dictionary. Based on this principle, Yuan and Tang [23] offered a sparse representation that used the set-to-set distance, which incorporated the spatial information into the sparse representation model using the set-to-set distance for accurate classification of HSIs. Liu et al. [25] proposed a neighboring filtering kernel related to a spectral-spatial kernel sparse representation that can measure spatial similarity by means of neighbor filtering in the kernel feature space, thus enhancing the performance of the classification of HSIs. Although spectral-spatial classification methods can achieve better performance than strictly spectral-based methods, traditional classification methods of HSIs tend to rely on manually crafted features and lack deep feature extraction, resulting in poor robustness and generalization of the classification model.

To realize automatic feature characterization, deep learning-based techniques have been widely applied in the classification of HSIs during the last few years [2], [29], [30], [31]. These studies focus on methods such as convolution neural networks (CNNs) [32], [33], recurrent neural networks (RNNs) [34], [35], and generative adversarial networks (GANs) [36]. Compared with traditional classification methods, deep learning-based methods can automatically extract low-level features (e.g., shape, geometry, and texture) and high-level semantic information through multilayer nonlinear transformations [37]. Given abundant samples, deep learning methods can effectively and accurately "learn" the relationships between input data and their corresponding labels (i.e., the conditional probability distribution of the data). Such methods can further distinguish objects by feature extraction of input data and decision boundary learning, in which the CNNs, RNNs, and graph convolution networks (GCNs) [38] are the common models. For example, the method proposed by Chen et al. [39] is perhaps the earliest work that introduced deep learning into the classification of HSIs, which stacked multiple autoencoder networks to extract deep spectral-spatial features from dimension-reduced HSIs generated by a PCA. Instead of stacking autoencoders, Chen et al. [33] employed CNNs to semantically extract spatial-spectral features by considering local contextual information of HSIs. Zhou et al. [40] input time series of each multichannel pixel and row vector of each patch into the LSTM model, respectively, to extract spectral-spatial features achieved outstanding performance for classification of HSIs; however, limited and unbalanced samples of HSIs pose a challenge for these models to learn the features and inherent patterns in the data, resulting in a decrease in both accuracy and generalization. To alleviate the dependence on labeled samples, Qin et al. [41] proposed a novel semi-supervised learning framework for classification of HSIs based on spectral-spatial GCN, which used the adjacency nodes in the graph to approximate the convolution while making full use of the spatial information of the current pixel. Some generative models, such as GAN and variational autoencoder (VAE), moreover, seek to generate additional samples by simulating the real data distribution, which can enhance the sample diversity, and mitigate issues associated with class sample imbalance, further improve the classification performance and robustness of classification models. Zhu et al. [42] introduced a spectral-spatial GAN for classification of HSIs, which uses adversarial training between generator and discriminator to simulate data distributions and generate high-quality samples, and then an embedded spectral-spatial classifier was used to extract spectral-spatial features from expanded samples, thereby enhancing the performance of the classification of HSIs. In order to further improve the performance of classification, Feng et al. [2] designed two generators to generate spectra by 1-D transposed convolutional network (1D-TCN), and spatial patches by 2D-TCN, respectively. The discriminator is determined to extract joint spatial-spectral features for accurate classification of HSIs. Although the above methods can obtain satisfactory performance in the classification of HSIs, existing efforts are mainly founded on the premise of a closed-set assumption, wherein the training and testing data are drawn from the same label and feature spaces; nevertheless, new land cover types are constantly emerging in real scenarios, resulting in an inability of fixed classification schemes to match them. Under this condition, traditional classification of HSIs prefers to classify the unknown class into one of the known classes, and the existence of open space (open-set) will inevitably increase the risk of misclassification. It is, therefore, of practical significance to design algorithms for open-set classification of HSIs, but relatively little attention has been devoted to this problem in recent years.

To meet the challenge of open-set recognition (i.e., OSR), some studies have explored open-set classification of HSIs. For example, Kang et al. [43] used the discriminative spectral-spatial feature extracted by CNN to construct a scalable prototype set to identify known samples and unknown samples. Sun et al. [44] constructed a feature learning framework based on reciprocal points learning to model extra-class space and then used a learnable dynamic threshold of known classes to distinguish unknown classes. Pal et al. [45] designed a residual 3-D convolutional block attention module to extract discriminative prototypes for the known classes and further presented an outlier calibration network based on meta-learning to distinguish known and unknown classes; moreover, researchers also concentrated on the OSR problem of natural images and achieved satisfactory results [46], [47], [48], [49], where generative-based methods were proven to perform better on unknown class recognition due to a decision boundary recognition ability with generated samples [44]. In view of this, Yue et al. [50] presented a spectral-spatial latent reconstruction learning method to enhance the learned feature representation by reconstructing the spectral and spatial features of the HSIs, so as to retain the spectral-spatial information useful for rejecting unknown classes and distinguishing known classes. Typically, generative adversarial models can effectively simulate the distribution of real samples through the adversarial process between generator

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and discriminator, which facilitates the accurate recognition of decision boundary between different classes. This is because the generator must consider the differences in real data distributions when generating samples, including differences between different classes. Pal et al. [51] offered a meta-based OSR method via a GAN to learn the finer separation between the closed and open spaces of HSIs. In this case, two GANs and a novel Antioverlap Latent "regularizer" were used to generate class-conditioned samples for both the closed and open spaces. Generally, existing OSR methods mainly aim to learn the distribution features (i.e., within-distribution) of known classes, and further find the decision boundary between "known and unknown," so as to reject samples that are out of the distribution (OOD); however, the significance of uncertainty that has been shown prevalent in the scope of the OOD problem, tends to be ignored in current open-set classification of HSIs [52], [53], [54].

Recently, evidential learning (EL) that incorporates the evidential theory into deep learning has been developed to quantify this uncertainty and has achieved promising results [55], [56]. Innovatively, the EL seeks to estimate classification uncertainty based on the collected classification evidence of different classes during model training. Different from mainstream Bayesian inference-based uncertainty estimation, EL-based methods require no complicated posterior probability estimation or sampling. For example, the concept of EL was introduced to quantify classification uncertainty in [55] for the first time, which proved the advancement of EL-based methods in uncertainty modeling. Inspired by Sensoy et al. [55], Bao et al. [56] used EL in open-set action recognition task while considering uncertainty calibration and deviation elimination simultaneously and realized satisfactory performance. Instead of focusing on the discrete classification problem, Amini et al. [57] introduced EL into the continuous regression problem to model the uncertainty by placing evidential priors over the Gaussian likelihood function and training the neural network to infer the hyperparameters of the evidential distribution. This provided useful guidelines for later uncertainty modeling research of the regression problem, such as [58], [59], [60]. Although the advantage of EL in uncertainty estimation can provide a new way for unknown class recognition, the EL still remains constrained by the inherent reliance of deep learning model training on a substantial quantity of high-quality samples; therefore, combining the EL with generative models for uncertainty modeling will make it possible to further improve the accuracy of open-set classification of HSIs.

In this study, we proposed a novel spectral-spatial EL (SSEL) network for open-set classification of HSIs by incorporating the evidential theory into the GAN framework. This solves the problem that arises when unclear decision boundaries result in low accuracies of open-set classification of HSIs under existing criteria. First, an improved GAN model is constructed to generate high-quality samples and extract spectral-spatial features for multiclass evidence acquisition, where a new classification module and adversarial loss are designed to make the decision boundary of "known-known" and "known-unknown" classes clearer. Then, the collected evidence is used to optimize the classification loss for better closed-set classification and uncertainty quantization. Finally, we recognize the known and unknown classes according to the results of closed-set classification and estimated uncertainty. Experimental results on three public datasets illustrate that the proposed SSEL significantly outperforms other existing advanced open-set classification methods of HSIs on recognition of unknown classes while keeping acceptable accuracy on known classes. The main contributions of this article can be summarized as follows.

1) We proposed a novel SSEL network based on improved GAN and evidential theory for open-set classification of HSIs.
2) We introduced the EL into the open-set classification task of HSIs for the first time, realizing high accuracy recognition of unknown classes while maintaining the classification accuracy of known classes.

The rest of this article is organized as follows. Section II describes the principle of the proposed methodology in detail. In Section III, the experimental setup is described, including datasets and training details. In Section IV, we discuss the experimental results obtained on three public datasets and provide related analysis. Finally, we present our conclusion in Section V.

II. Methodology

In this section, the proposed open-set classification framework of HSIs will be described in detail. First, we briefly introduce the overall structure of the framework. Second, the evidence collection module combining GAN and evidential theory is introduced. Finally, we present a novel open-set classification module of HSIs based on uncertainty estimation.

A. Proposed Framework

Fig. 1 shows the overview of the proposed processing scheme, which consists of two key parts: 1) evidence collection based on improved GAN and evidential theory and 2) open-set classification of HSIs based on uncertainty estimation. In order to realize accurate open-set classification of HSIs, there are two conditions that need to be prepared ahead of: first, the result of a closed-set classification, and second, multiclass evidence for uncertainty estimation; thus, an improved GAN model is proposed in this study. In the improved GAN model, the generator is used to expand high-quality samples, and the discriminator is redesigned to conduct multiclass evidence collection and closed-set classification. To be specific, the extra label information is added to randomize the noise, which stabilizes and boosts the generator to generate samples of specific types. In addition, to generate samples closer to the distribution of real samples, a domain adaptation (DA) technique is introduced to constrain the similarity between generated and real samples and further guide the training process of the model. More importantly, a novel classification loss based on evidential theory was added to the discriminator to guide the evidence collection and improve the accuracy of the closed-set classification process of the discriminator. Correspondingly, the adversarial loss of the traditional GANs

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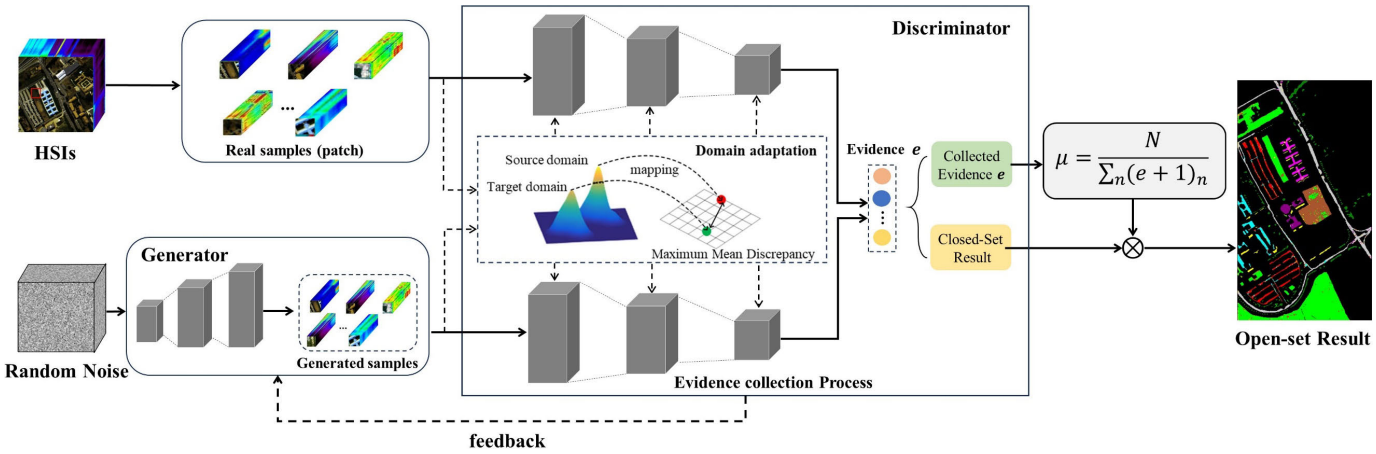


Fig. 1. Flowchart of the proposed open-set classification method of HSIs.

is also redesigned to better guide the training of the GAN model, where the goal of the generator loss is to guide the generator to generate samples belonging to the class that are consistent with the label and the discriminator loss aims to guide discriminators to distinguish real and fake samples as accurately as possible.

Different from closed-set classification, there will be unknown classes emerging in test sets; therefore, the collected evidence is also used to quantify the uncertainty, and the unknown samples will be recognized by combining the results of the closed-set classification and the estimated uncertainty.

B. Evidence Collection Module Based on GAN and Evidential Theory

To extract robust spectral-spatial features for multiclass evidence collection, we proposed a novel multiclass spectral-spatial GAN that incorporates evidential theory and improves the model for both the generator and discriminator.

For the generator (G), the improved GAN considers both extra label information and DA to improve the quality of generated samples. Assume a 3-D random noise cube $r \in \mathbb{N}^{5\times5\times c}$, where c denotes the number of spectral channels. To stabilize the generator and avoid potential model collapse, the class label $y \in \Upsilon^{5\times5\times N}$ is added to r to get the input of the generator $z \in \mathcal{R}^{5\times5\times C}$, where N is the number of classes, and the value of C equals the sum of c and N. The generated samples, furthermore, often differ greatly from the real samples. To further ensure the quality of the generated samples, we also introduced a DA technique that can reduce the discrepancy between two separate domains (source domain and target domain) to enhance the training process and constrain the similarity between generated and real samples. To better measure the difference between the distribution of the source domain (real samples) and the target domain (generated samples), the maximum mean discrepancy (MMD) loss function was used to calculate the distance between the probability distributions of the source domain and the target domain. Specifically, given the real samples $X \in \mathbb{R}^{5\times5\times D}$ with the distribution P, and the input of generator z, the generator aims to map the z from the model space to the real sample space $G(z) \in \mathbb{R}^{5\times5\times D}$ with the distribution $\mathbb{Q}$. Then, the MMD loss uses the kernel function k to measure the difference between the two samples. The distance of the square MMD can be formulated as follows:

$$\begin{aligned} D_k^2(\mathbb{P},\mathbb{Q}) &= \left\|\mu_{\mathbb{P}} - \mu_{\mathbb{Q}}\right\|_{\mathcal{H}}^2 \\ &= \mathbb{E}_{\mathbb{P}}\left[k\left(x,x'\right)\right] - 2\mathbb{E}_{\mathbb{P},\mathbb{Q}}\left[k(x,g)\right] \\ &\quad + \mathbb{E}_{\mathbb{Q}}\left[k\left(g,g'\right)\right]. \end{aligned} \tag{1}$$

Here, x and x' denote two real samples, and g and g' are two samples from the generator. In addition, E is the operator to calculate expectation. Here, $k(x, g)$ is used to measure the similarity between two samples, and $D_k^2(\mathbb{P},\mathbb{Q})$ will be zero when the distribution of generated samples and real samples are the same. The MMD loss was, therefore, added to the objective function of the generator to stabilize the training process. The objective function has the following formulation:

$$\mathcal{L}_{G_\text{ada}} = \mathbb{E}_{\mathbb{P}}\left[k\left(x,x'\right)\right] - 2\mathbb{E}_{\mathbb{P},\mathbb{Q}}\left[k(x,g)\right] + \mathbb{E}_{\mathbb{Q}}\left[k\left(g,g'\right)\right]. \tag{2}$$

On the one hand, the existence of MMD loss will force generated samples closer to the distribution of real samples by maximizing $\mathbb{E}_{\mathbb{P},\mathbb{Q}}[k(x, g)]$. On the other hand, the intraclass similarity of generated samples will also be improved by maximizing $\mathbb{E}_{\mathbb{Q}}[k(g, g')]$, which will help to determine a clear decision boundary.

In discriminator D, the inputs contain the real samples $X \in \mathbb{R}^{5\times5\times D}$, generated samples $G(z)$ and corresponding labels y. Different from the discriminator of traditional GANs that only distinguishes whether the input sample is real or fake, the improved discriminator aims to collect multiclass evidence for both closed-set classification and uncertainty estimation. Specifically, the spectral-spatial features of the input samples are first extracted, and then the extracted features are input to the classifier for evidence integration. To collect sufficient evidence to support accurate classification, a novel classification loss function called evidence-based loss is designed to guide the training of the discriminator. Specifically, according to the EL theory proposed by [55], once given the sample X_i for N-class classification, the standard output of a classification network is the parameter set of a categorical distribution, which can be replaced with the

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distribution parameters α of a Dirichlet density $D(p, \alpha)$. Also, the cross-entropy loss to be minimized for classification can eventually be expressed by the following:

$$\mathcal{L}_{\text{cla}}^{(j)}(\Theta) = \sum_{i=1}^{N} y_i^{(j)} \left(\log S^{(j)} - \log\left(e_i^{(j)} + 1\right)\right) \tag{3}$$

where Θ refers to network parameters and e is the evidence vector predicted by the network, which is a measure of the support derived from data in favor of a sample to fall into a certain class. The evidence e can be expressed as follows:

$$e = g(f(X|\Theta)) \tag{4}$$

where f is the output of a network parameterized by Θ and g is the activation function to keep evidence e nonnegative, and S denotes the Dirichlet strength, which is defined as follows:

$$S = \sum_{i=1}^{N} (e_i + 1). \tag{5}$$

Based on the Dempster-Shafer theory [61] and subjective logic [62], the distribution parameters α and those learned (e) satisfy that $\alpha = e + 1$. Once the parameters α is given, the expected probability of the ith class is the mean of the corresponding Dirichlet distribution and computed as

$$P_i = \frac{\alpha_i}{S}. \tag{6}$$

Minimizing the loss function $\mathcal{L}_{\text{cla}}$ is accomplished by maximizing the collection of evidence to support the correct classification. To further improve the discriminative ability to distinguish between real and fake samples, the DA technique is also added to the discriminator, and the objective function is as follows:

$$\mathcal{L}_{D_\text{ada}} = \mathbb{E}_{\mathbb{P}}\left[k\left(x, x'\right)\right] - 2\mathbb{E}_{\mathbb{P},\mathbb{Q}}\left[k(x, g)\right] + \mathbb{E}_{\mathbb{Q}}\left[k\left(g, g'\right)\right]. \tag{7}$$

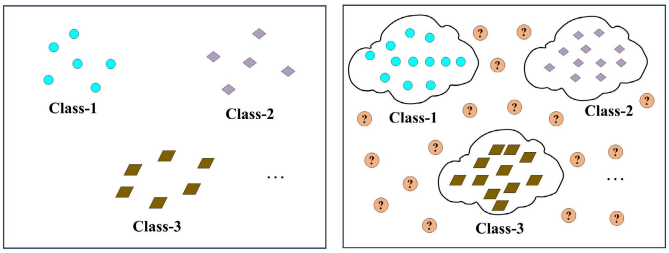
To maximize the loss function, the discriminator aims to reduce $\mathbb{E}_{\mathbb{P},\mathbb{Q}}[k(x, g)]$ and that forces the generated samples away from the real samples, so as to better distinguish them.

Correspondingly, the new adversarial objective function was redefined to drive the training of the model, and the formula is shown below

$$\mathcal{L}_{D_\text{adv}} = \sum_{i=1}^{N} \mathcal{L}_{\text{cla}}(D(X_i), y_i + \mathcal{L}_{\text{cla}}(D(G(z_i)), y_{G_i}) \tag{8}$$

$$\mathcal{L}_{G_\text{adv}} = \sum_{i=1}^{N} \mathcal{L}_{\text{cla}}(D(G(z_i)), y_i). \tag{9}$$

As can be seen from (8), the adversarial loss of the discriminator still includes two parts, as do traditional GANs: the classification loss of real samples $\mathcal{L}_{\text{cla}}(D(X_i), y_i)$ and the classification loss of generated samples $\mathcal{L}_{\text{cla}}(D(G(z_i)), y_{G_i})$. The former drives the discriminator to assign a higher probability to the class corresponding to the sample. In contrast, the latter expects the discriminator to classify the generated samples into none of the classes; that is, the class probability of the generated samples should be $[0, 0, 0, \ldots, 0]$. Correspondingly, the label y_G should be $[0, 0, 0, \ldots, 0]$. According to the formula (6), the expected probability P_i of the ith class is, however, greater than 0; thus, a novel label $y_G = [(1/N), (1/N), (1/N), \ldots, (1/N)]$ is used to replace $y_G = [0, 0, 0, \ldots, 0]$ in our study, which approximately assigns the generated samples as none of the classes. The updated labels y_G can, moreover, help the model to better determine the decision boundary between unknown and known classes. For the adversarial loss of the generator, the goal of which is to motivate the generator to generate samples with a distribution that is closer to the real samples. The redefined adversarial objective function of L_{D_adv} and L_{G_adv} are generally adversarial with each other to improve the generative ability of high-quality samples and the discriminative ability to distinguish between real and fake samples.



(a) (b)

Fig. 2. Schematic of sample generation (a) original samples and (b) amplified samples and decision boundary.

In general, the multiclass spectral-spatial GAN offers three improvements over traditional GANs used for classification. First, MMD-based DA is introduced to stabilize the training process and generate high-quality samples. Second, evidence-based classification loss is used to guide the training of the discriminator, which collects sufficient evidence to support accurate closed-set classification and uncertainty estimation. Third, a novel adversarial objective function is designed to drive the training process and boost classification performance. The final loss function of the discriminator and generator can be formulated as

$$\mathcal{L}_D = \mathcal{L}_{D_\text{adv}} + \mathcal{L}_{D_\text{ada}} \tag{10}$$

$$\mathcal{L}_G = \mathcal{L}_{G_\text{adv}} + \mathcal{L}_{G_\text{ada}} \tag{11}$$

and the final objective function of the GAN is represented as

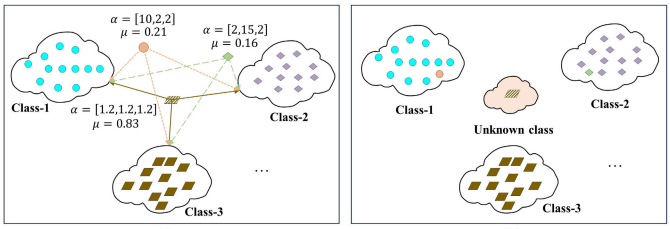
$$\mathcal{L}_{\text{GAN}}(G, D) = \min_G \max_D (\mathcal{L}_G + \mathcal{L}_D). \tag{12}$$

Thanks to the improved GAN model, a variety of high-quality samples will be generated and the decision boundary between the classes will also be further clarified during model training, as shown in Fig. 2. This will help to improve the accuracy of both closed-set and open-set classification of HSIs.

C. Open-Set Classification of HSIs Based on Uncertainty Estimation

To recognize unknown samples in the test set, we employed an open-set classification method of HSIs based on uncertainty estimation by combining closed-set classification results with collected evidence. Given the test data, uncertainty represents the uncertainty degree for the prediction results of the

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(a) (b)

Fig. 3. Schematic of open-set classification of HSIs based on uncertainty estimation. (a) Closed-set classification results and collected evidence. (b) Open-set classification results.

model. In general, the higher the uncertainty, the greater the probability that indicates that the classification result belongs to the unknown class. Given a sample X, the uncertainty will be low when collected evidence is concentrated in a certain class. In contrast, the uncertainty will be high when collected evidence makes it hard to recognize the specific sample class; therefore, the uncertainty can be used to discriminate between the known and unknown classes in practice. The plots in Fig. 3 provide the schematic of open-set classification of HSIs based on uncertainty estimation. As can be seen from Fig. 3(a), the samples will be first assigned a specific class after being input to the improved GAN model, and the corresponding evidence e on each class will also be given at the same time. According to EL theory, the predictive uncertainty μ of the samples can be calculated based on the collected evidence as follows:

$$\mu = \frac{N}{S} \tag{13}$$

where N denotes the number of the classes, S is the Dirichlet strength, and $S = \sum_{i=1}^{N}(e_i + 1)$, as shown in formula (5). In this way, the uncertainty is less likely to suffer from over-confidence common in traditional methods (e.g., softmax), because it avoids merely regarding the probability of labeled class as uncertainty but additionally taking into account the comparative values against other classes. Then, the discriminative coefficient ω is used to distinguish known and unknown classes, which can be calculated based on the uncertainty threshold δ

$$\omega = \begin{cases} 1, & \text{if } \mu < \delta \\ 0, & \text{if } \mu \geq \delta \end{cases} \tag{14}$$

where "1" denotes the known classes, and "0" represents the unknown class. Finally, the result of open-set classification $\mathbb{C}_{\text{open}}$ of HSIs can be obtained by combining the closed-set classification $\mathbb{C}_{\text{closed}}$ result with the discriminative coefficient ω [see Fig. 3(b)], which can now be expressed as

$$\mathbb{C}_{\text{open}} = \omega \times \mathbb{C}_{\text{closed}}. \tag{15}$$

III. Experiments

In this section, we describe the open-set datasets used in our study while introducing the implementation details of the experiments conducted.

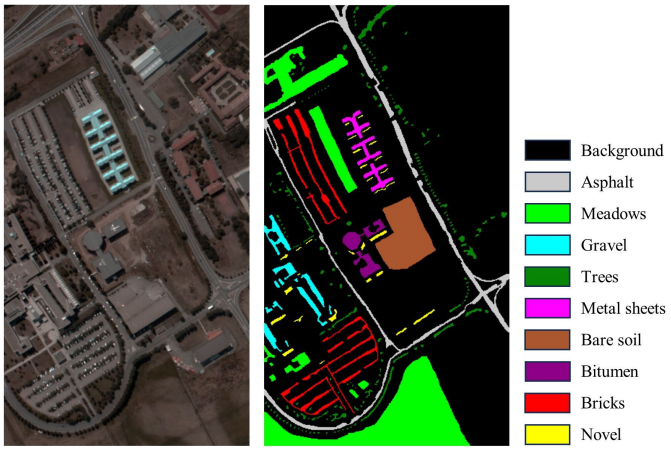


Fig. 4. Pseudo-color composite image of the University of Pavia dataset, the corresponding ground truth, and legend.

A. Dataset Description

To verify the effectiveness and advancement of the proposed method, three public hyperspectral datasets, specifically the University of Pavia dataset, the Salinas dataset, and the WHU-Hi-LongKou dataset, are used in this study. For each dataset, the open-set environments are created, and the detailed open-set environments of these three datasets are listed in Table I.

1) University of Pavia Dataset: The University of Pavia dataset was captured by a reflective optics system imaging spectrometer (ROSIS) optical sensor in the area around the University of Pavia, Italy. The image size is 610 × 340, with a spatial resolution of 1.3 m per pixel. The ROSIS sensor provides 115 spectral bands with wavelengths ranging from 0.43 to 0.86 μm. In practice, twelve bands were excluded before conducting the experiment since these bands are strongly influenced by noise. The dataset contains a total of 9 object classes, including asphalt, meadows, gravel, trees, painted metal sheets, bare soil, bitumen, self-blocking bricks, and shadows. To validate the performance of our proposed open-set classification methods of HSIs, the last class "shadows," is selected as the unknown class, and the remaining eight classes are selected as known classes. The three parts of Fig. 4 are the pseudo-color composite image, the ground truth image, and the legend of the dataset.

2) Salinas Dataset: The Salinas dataset was acquired by an airborne visible/infrared imaging spectrometer (AVIRIS) over Salinas Valley, California. The image size is 512 × 217, with a spatial resolution of 3.7 m per pixel. It contains 224 spectral bands ranging in wavelengths from 0.4 to 2.5 μm. Similar to the Paiva University dataset, 20 bands that are absorbed by water were eliminated, and only 204 bands were retained. The original dataset contains a total of 16 object classes. In our experiment, the last two classes are selected as the unknown classes, and the remaining 14 classes are selected as the known classes. In Fig. 5, we present the pseudo-color composite image, the ground truth image, and the corresponding color legend of the Salinas dataset.

3) WHU-Hi-LongKou Dataset: The WHU-Hi-LongKou dataset was acquired in the area around LongKou, Hubei, China, with an 8-mm focal length Headwall Nano-Hyperspec

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TABLE I
DETAILED OPEN-SET ENVIRONMENTS OF THE THREE DATASETS

	University of Pavia			Salinas			WHU-Hi-LongKou		
ID	Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test
1	Asphalt	50/200	6581/6431	Weeds_1	50/200	1959/1809	Corn	50/200	34461/34311
2	Meadows	50/200	18599/18449	Weeds_2	50/200	3676/3526	Cotton	50/200	8324/8174
3	Gravel	50/200	2049/1899	Fallow	50/200	1926/1776	Sesame	50/200	2981/2831
4	Trees	50/200	3014/2864	Fallow-P	50/200	1344/1194	B-L Soybean	50/200	63162/63012
5	Metal sheets	50/200	1295/1145	Fallow-S	50/200	2628/2478	N-L Soybean	50/200	4101/3951
6	Bare soil	50/200	4979/4829	Stubble	50/200	3909/3759	Rice	50/200	11804/11654
7	Bitumen	50/200	1280/1130	Celery	50/200	3529/3379	Water	50/200	70006/69856
8	Bricks	50/200	3632/3482	Grapes	50/200	11221/11071	Roads-Houses	50/200	7074/6924
9	Novel	0	947	Soil	50/200	6153/6003	Novel	0	5229
10	Total	400/1600	42376/41176	Corn	50/200	3228/3078	Total	400/1600	207142/205942
11				Lettuce-4wk	50/200	1018/868			
12				Lettuce-5wk	50/200	1877/1727			
13				Lettuce-6wk	50/200	866/716			
14				Lettuce-7wk	50/200	1020/870			
15				Novel	0	9075			
				Total	700/2800	53429/51329			

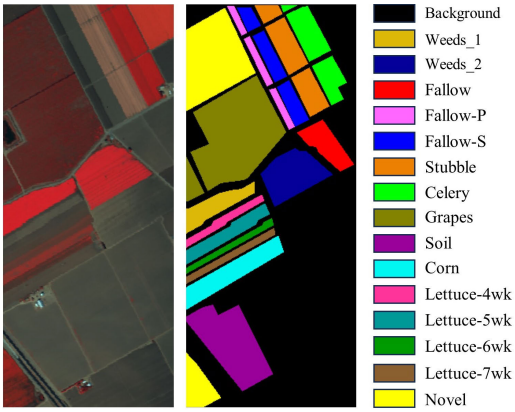


Fig. 5. Pseudo-color composite image of the Salinas dataset, the corresponding ground truth, and the legend.

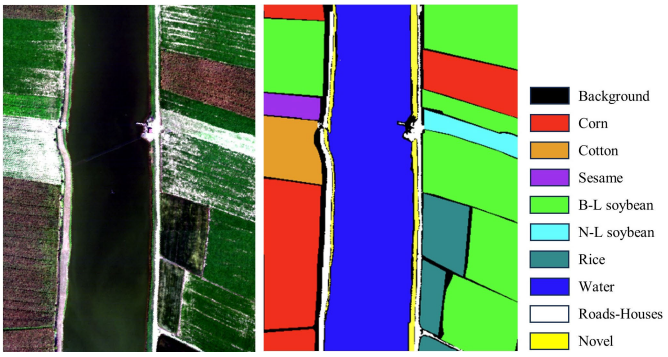


Fig. 6. Pseudo-color composite image of the WHU-Hi-LongKou dataset, the corresponding ground truth, and legend.

imaging sensor equipped on a DJI Matrice 600 Pro UAV platform [63]. The image size is 550 × 400, there are 270 bands from 400 to 1000 nm, and the spatial resolution is about 0.463 m. The study area contains six crop species: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. Besides, there are a total of 9 object classes defined in the dataset. To validate the performance of our method, the last class is selected as the unknown class, and the remaining eight classes are selected as known classes. An overview of the dataset is provided in Fig. 6

For each dataset, we selected different sample proportions (50 samples per class and 200 samples per class, i.e., $S = 50$ and $S = 200$, S denotes the number of samples) to train the model. The rest of the samples from each image were used as the test dataset, and the background pixels in the dataset were masked.

B. Experimental Settings

In order to extract spectral-spatial features for classification of HSIs and explore the influence of patch size on classification accuracy, various patch sizes of 1 × 1, 3 × 3, 5 × 5, and 7 × 7 were used to conduct experiments on the datasets. The patch size of 5 × 5 is finally selected as the optimal size. Meanwhile, to comprehensively verify the advantage of the proposed method, several state-of-the-art open-set classification methods of HSIs, such as multitask deep learning method for open world (MDL4OW) [64], and spectral-spatial latent recognition (SSLR) [50] were employed for comparison. We also selected two outstanding closed-set classification methods of HSIs for comparison, namely 3-D convolutional neural network (3DCNN) [65] and spectral-spatial residual network (SSRN) [66]. For the above methods, the related parameter settings are consistent with those in our study. To compare the accuracy of the proposed method with other benchmark methods for open-set classification of HSIs, we used the overall accuracy (OA_o), the average accuracy (AA_o), and the kappa coefficient (K_o). In addition, we also evaluated the accuracies under a closed-set environment using the above evaluation indices (OA_c, AA_c, and K_c), to verify the improved performance of the proposed method.

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TABLE II
DETAILED STRUCTURE AND PARAMETERS OF THE IMPROVED GAN

Nets	Layer	Convolution	Stride	Padding	Activation
G	1	P×P×128	1	SAME	ReLU
	2	P×P×256	1	SAME	ReLU
	3	P×P×256	1	SAME	ReLU
	4	P×P×104/204/270	1	SAME	ReLU
D	1	P×P×50	1	SAME	ReLU
	2	P×P×100	1	SAME	ReLU
	3	P×P×200	1	SAME	ReLU
	4	P×P×100	1	SAME	ReLU
	5	P×P×50	1	SAME	ReLU
	6	FC:M×N			

For the improved GAN model, the main network structure and parameters used in this study are shown in Table II. In Table II, G and D refer to the generator and discriminator, respectively. P denotes the patch size. N represents the number of classes, and M denotes the number of extracted spectral-spatial features. The initial learning rate of the generator and discriminator were set to 0.00017 and 0.00005, respectively, and an Adam optimizer was used to adaptively adjust the learning rate in the training phase. The number of epochs was set as 3000. The training process of the improved GAN model was, moreover, completed on a computer consisting of 11th Gen Intel[1] Core[2] i9-11900K, RAM (128 GB), and GPU (Nvidia GeForce RTX 3060 with 12 GB). In addition, the deep learning models used in this study all employed the TensorFlow framework, and the version of Python is 3.7. For the discriminative coefficient, the uncertainty threshold δ was set to 0.6.

IV. RESULTS AND DISCUSSION

A. Results and Analysis

1) Classification Results for the University of Pavia Datase: The first experiment was conducted on the University of Pavia dataset, and Table III presents the objective accuracies of the different classification methods of HSIs under different sample proportions. In general, it is clear to see that our proposed method achieves the best performance under an open-set environment compared with the advanced closed-set classification methods of HSIs in terms of OA_o, AA_o, and K_o. Specifically, compared with the 3DCNN method, when $S = 50$, the OA_o, AA_o, and K_o of the proposed SSEL increase by 2.73%, 9.76%, and 3.81%, respectively. Compared with the SSRN method, when $S = 50$, the OA_o, AA_o, and K_o of the proposed SSEL increase by 1.03%, 8.30%, and 1.62%, respectively. Under 200 samples context, compared with the 3DCNN method, the OA_o, AA_o, and K_o of the proposed SSEL increase by 2.75%, 9.47%, and 3.79%, respectively. And compared with the SSRN method, the OA_o, AA_o, and K_o of the SSEL method increase by 1.40%, 6.97%, and 1.94%, respectively; moreover, the above closed-set classification methods cannot deal with the unknown class due to the lack of samples of the unknown classes. In contrast, the proposed SSEL method achieves satisfactory performance on unknown class recognition with an accuracy of 91.34% and 95.25% under different sample proportions while maintaining competitive accuracy on the known classes.

From Table III, we can see that the proposed method also produces competitive accuracies under both open-set and closed-set environments compared with the state-of-the-art open-set classification methods of HSIs. Specifically, compared with the MDL4OW method, when $S = 50$, the OA_o, AA_o, and K_o of our proposed SSEL are increased by 1.15%, 4.90%, and 1.70%, respectively, and the OA_c and K_c are increased by 0.05% and 0.24%. Also, compared with the recently proposed SSLR method for open-set classification of HSIs, our proposed SSEL still achieves 0.50%, 4.03%, and 0.84% improvements in OA_o, AA_o, and K_o. When $S = 200$, compared with the MDL4OW method, the OA_o, AA_o, and K_o of the SSEL are increased by 1.37%, 4.85%, and 1.82%, respectively, and the OA_c and K_c are increased by 0.27% and 0.32%. Compared with the SSLR method, the SSEL method still achieves 2.43%, 5.21%, and 3.30% improvements in OA_o, AA_o, and K_o, while the OA_c, AA_c, and K_c are increased by 1.48%, 0.28%, and 2.04%, respectively. Our method, moreover, greatly outperforms the above advanced open-set classification methods of HSIs for unknown class recognition, which achieves 49.41%, 43.93%, 49.32%, and 44.56% improvement under different sample proportions, respectively.

For illustrative purposes, we show the classification maps of the different methods under different sample proportions for the University of Pavia dataset in Fig. 7. We can see that the proposed method can effectively recognize the unknown class, which mainly benefits from the evidence collection and accurate decision boundary recognition. Among the four comparative experiments, only two open-set classification methods (MDL4OW and SSLR) of HSIs recognized the unknown class but with poor performance, while two closed-set methods directly classified the unknown classes into known classes due to the lack of prior knowledge of the unknown classes. The results of Fig. 7, moreover, also demonstrate that the proposed SSEL method still achieves outstanding performance for the known classes.

2) Classification Results of the Salinas Dataset: The second experiment was performed on the Salinas dataset, and we also report the accuracies of the proposed SSEL method and the comparison methods for the various classes in Table IV. From this table, we can see that our proposed method achieves the best performance under an open-set environment compared with the advanced closed-set classification methods of HSIs. Under 50 samples context, compared with the 3DCNN method, the OA_o, AA_o, and K_o of the proposed SSEL increased by 11.91%, 4.92%, and 13.69%, respectively. Compared with the SSRN method, the OA_o, AA_o, and K_o of the SSEL method increase by 9.50%, 2.38%, and 10.84%. As for the 200 samples setting, compared with the 3DCNN

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[2]Trademarked.

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TABLE III
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR BOTH KNOWN AND UNKNOWN CLASSES ON THE UNIVERSITY OF PAVIA DATASET

Class No.	S=50					S=200				
	3DCNN	SSRN	MDL4OW	SSLR	SSEL	3DCNN	SSRN	MDL4OW	SSLR	SSEL
1	87.54	85.80	89.65	88.61	85.39	90.29	95.17	91.74	89.65	93.31
2	89.95	92.54	90.97	90.95	93.74	93.88	90.84	92.18	93.99	95.64
3	84.09	81.83	83.94	76.61	93.02	84.42	82.38	87.33	89.99	91.40
4	95.78	97.62	97.94	97.68	92.58	97.11	97.94	98.27	98.24	96.44
5	99.68	99.76	95.46	94.50	86.02	99.91	100.00	84.24	92.71	86.20
6	88.68	88.80	89.62	91.51	81.11	88.98	98.54	97.51	81.35	87.14
7	90.41	95.28	92.26	95.56	94.55	97.88	99.32	96.17	94.59	96.37
8	84.20	91.85	82.05	91.17	90.42	87.82	98.52	88.48	87.40	83.69
Novel	0.00	0.00	42.23	47.41	**91.34**	0.00	0.00	45.93	50.69	**95.25**
OA_c	89.35	**91.09**	90.16	90.70	90.21	92.24	**93.69**	92.50	91.29	92.77
AA_c	90.04	**91.69**	90.24	90.57	89.61	92.54	**95.34**	91.99	90.99	91.27
K_c	85.91	**88.17**	86.97	87.70	87.21	89.59	**91.50**	90.08	88.36	90.40
OA_o	87.51	89.21	89.09	89.74	**90.24**	90.07	91.42	91.45	90.39	**92.82**
AA_o	80.04	81.50	84.90	85.77	**89.80**	82.25	84.75	86.87	86.51	**91.72**
K_o	83.61	85.80	85.72	86.58	**87.42**	86.82	88.67	88.79	87.31	**90.61**

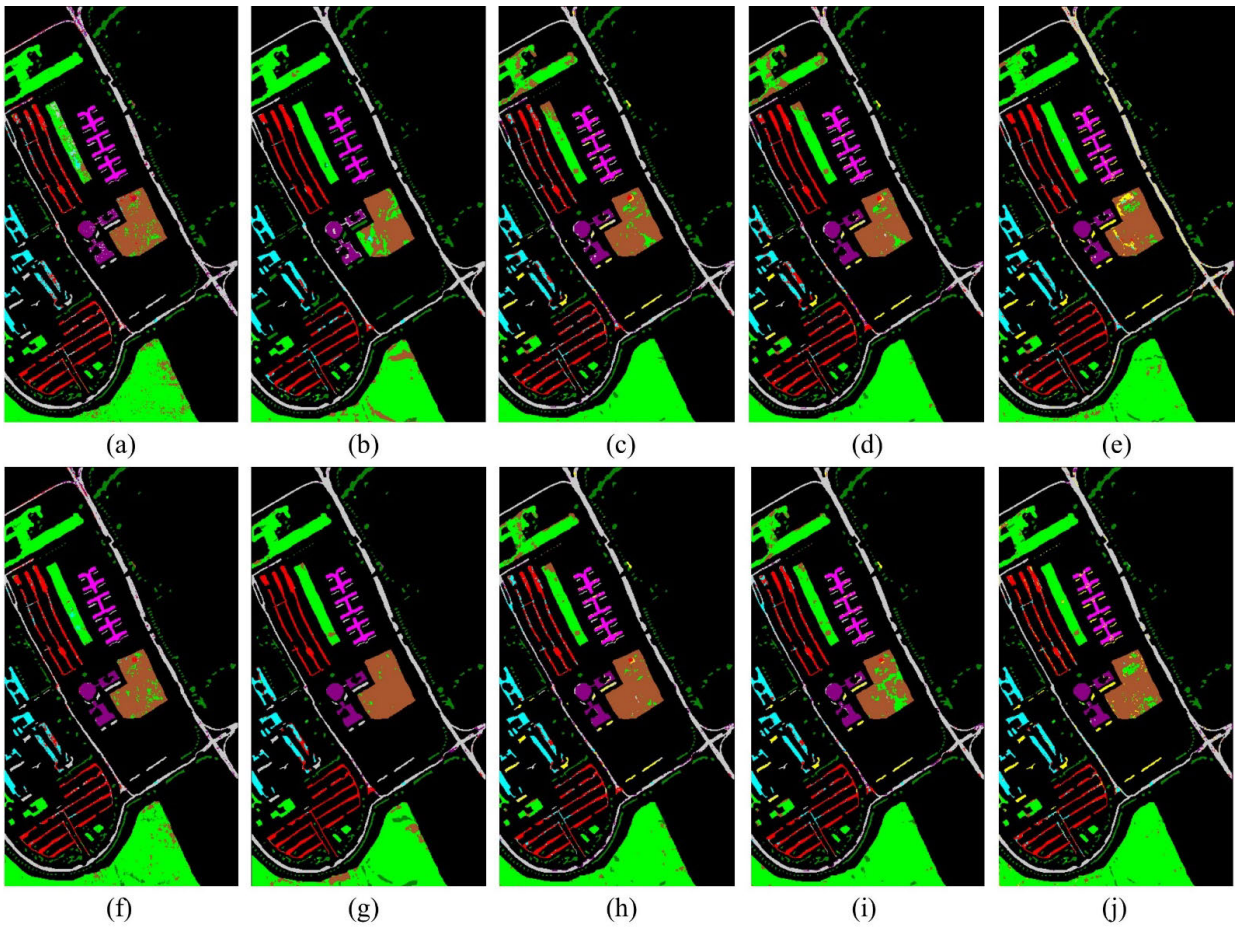


Fig. 7. Classification results of different methods for the University of Pavia dataset. (a) 3DCNN ($S = 50$). (b) SSRN ($S = 50$). (c) MDL4OW ($S = 50$). (d) SSLR ($S = 50$). (e) SSEL ($S = 50$). (f) 3DCNN ($S = 200$). (g) SSRN ($S = 200$). (h) MDL4OW ($S = 200$). (i) SSLR ($S = 200$). (j) SSEL ($S = 200$).

method, the OA_o, AA_o, and K_o of the SSEL increased by 13.15%, 4.55%, and 15.00%, respectively. Compared with the SSRN method, the OA_o, AA_o, and K_o of the SSEL method increase by 11.18%, 2.68%, and 12.69%. Meanwhile, the 3DCNN and SSRN methods fail to recognize the unknown class properly. For the unknown classes, our

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TABLE IV
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR BOTH KNOWN AND UNKNOWN CLASSES ON THE SALINAS DATASET

Class No.	S=50					S=200				
	3DCNN	SSRN	MDL4OW	SSLR	SSEL	3DCNN	SSRN	MDL4OW	SSLR	SSEL
1	98.91	90.23	99.43	99.01	100.00	99.89	98.31	98.37	99.10	99.66
2	99.01	100.00	99.83	99.27	99.83	98.55	100.00	99.94	100.00	99.55
3	84.74	90.99	66.61	96.12	99.72	85.25	94.43	95.89	97.92	98.25
4	99.66	99.57	99.50	99.08	99.83	99.41	99.86	99.66	99.33	99.99
5	91.73	99.33	98.31	97.71	98.02	92.21	99.40	98.22	98.47	97.86
6	98.82	100.00	100.00	100.00	97.82	99.60	100.00	99.63	99.97	100.00
7	98.51	99.94	99.73	99.23	98.73	99.82	99.78	98.61	96.69	98.52
8	96.09	96.27	94.25	85.62	94.27	96.66	98.17	94.52	90.08	94.10
9	98.51	99.94	99.93	99.53	99.42	97.61	99.94	99.47	99.27	99.92
10	80.97	97.62	84.81	85.23	62.64	94.34	95.12	76.52	87.60	80.75
11	94.10	99.44	97.79	95.22	96.62	95.28	99.25	88.58	94.06	80.02
12	97.28	100.00	99.00	99.38	99.94	99.59	100.00	99.94	99.94	99.94
13	99.43	99.78	99.44	99.32	99.29	100.00	99.34	99.15	99.15	99.71
14	96.12	98.88	96.52	96.13	89.20	96.36	98.88	94.66	94.19	95.35
Novel	0.00	0.00	50.15	56.34	**72.44**	0.00	0.00	58.35	62.68	**80.60**
OA_c	95.57	**97.95**	95.64	94.41	94.96	96.91	**98.74**	95.92	95.64	96.14
AA_c	95.28	**98.00**	95.37	96.49	95.38	96.76	**98.75**	95.94	96.84	95.89
K_c	94.94	**97.68**	95.04	93.67	94.28	96.47	**98.57**	95.37	95.06	95.61
OA_o	79.12	81.53	87.70	87.76	**91.03**	80.22	82.19	89.36	89.88	**93.37**
AA_o	88.93	91.47	92.36	93.81	**93.85**	90.30	92.17	93.43	94.56	**94.85**
K_o	76.30	79.15	86.37	86.46	**89.99**	77.58	79.89	88.18	88.77	**92.58**

proposed method obtains the highest accuracy of 72.44% and 80.60% under different sample proportions. Additionally, our proposed method also achieves satisfactory accuracy under the closed-set environment compared with the other methods.

From Table IV, we can also see that the proposed SSEL method achieved the best performance under the open-set environment compared with the advanced open-set classification methods of HSIs. Specifically, compared with the MDL4OW method, when $S = 50$, the OA_o, AA_o, and K_o of the SSEL increase by 3.33%, 1.49%, and 3.62%, respectively. Also, when $S = 50$, the OA_o, AA_o, and K_o of the SSEL increase by 3.27%, 0.04%, and 3.53% compared with the SSLR method. Under 200 samples context, compared with the MDL4OW method, the OA_o, AA_o, and K_o of the SSEL increase by 4.01%, 1.42%, and 4.40%, respectively. And compared with the SSLR method, the OA_o, AA_o, and K_o are increased by 3.49%, 0.29%, and 3.81%, and the OA_c, and K_c. are increased by 0.50% and 0.55%; furthermore, our method also achieves better performance than the above advanced open-set classification methods of HSIs for unknown class recognition with an improvement of 22.29%, 16.10%, 22.25%, and 17.92%, respectively.

For illustrative purposes, we show the classifications maps of the different methods for the Salinas dataset in Fig. 8. The classification maps demonstrate that the SSEL method also achieves the best performance for recognition of unknown classes on the Salinas dataset, and similarly, to the University of Pavia dataset, two closed-set classification methods of HSIs still fail to recognize unknown classes and classify them as known classes, which is mainly because the closed-set method is based on the assumption that all classes are known, resulting in the model being unable to recognize unknown classes. Although two open-set classification methods can recognize the unknown classes to a certain extent, the performance is not satisfactory.

3) Classification Results of the WHU-Hi-LongKou Dataset: In the third experiment, we applied the proposed method and comparison methods to the WHU-Hi-LongKou dataset. In Table V, we show the accuracies obtained with 50 and 200 training samples per class. Under 50 samples context, the proposed SSEL achieves the best classification in terms of three evaluation metrics (94.56%, 91.56%, and 92.53%) under an open-set environment while obtaining competitive closed-set classification performance. Besides, the proposed SSEL method also achieves the best performance on unknown class recognition with an accuracy of 76.44% compared with other classification methods. Under 200 samples setting, the best open-set classification performance is also obtained by the proposed SSEL method, which achieves the accuracy of 96.31%, 92.89%, and 95.18% in terms of three evaluation metrics. When it comes to unknown class recognition, the proposed method significantly outperformed comparison methods, which achieves 27.53% and 36.95% improvement over other open-set classification methods, respectively.

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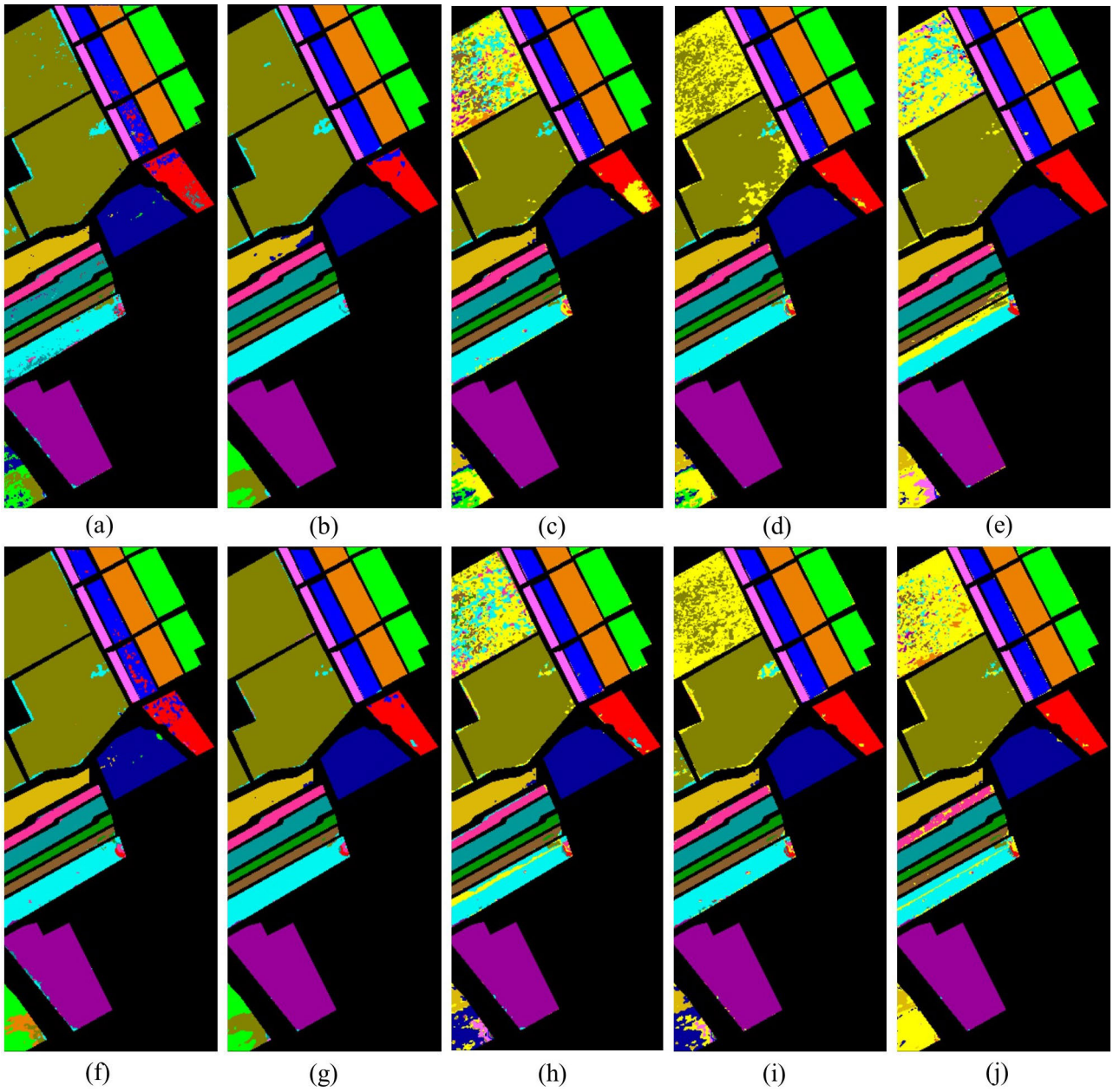


Fig. 8. Classification results of different methods for the Salinas dataset. (a) 3DCNN ($S = 50$). (b) SSRN ($S = 50$). (c) MDL4OW ($S = 50$). (d) SSLR ($S = 50$). (e) SSEL ($S = 50$). (f) 3DCNN ($S = 200$). (g) SSRN ($S = 200$). (h) MDL4OW ($S = 200$). (i) SSLR ($S = 200$). (j) SSEL ($S = 200$).

For illustrative purposes, we show the classification maps of the different methods for the WHU-Hi-LongKou dataset in Fig. 9. We can see that most unknown class instances are misclassified as known classes for comparison methods, especially the closed-set classification methods. In contrast, our proposed SSEL method successfully recognized unknown classes while maintaining high accuracy on the known classes, which further demonstrates the effectiveness of the proposed method.

B. Effectiveness Analysis of DA in the Improved GAN Model

To verify the effectiveness of the DA strategy, we performed comparisons between the GAN model with and without the DA module on three datasets. The relevant accuracies and results of the comparison are shown in Table VI. Table VI shows that the used DA module plays a crucial role in improving classification accuracy.

For the University of Pavia dataset, when the DA module was adopted, the accuracy of the model increased in the open-set environment under different sample proportions. This is mainly due to the DA module forcing the generated sample closer to the real one, and further, the intraclass similarity and interclass diversity of generated samples will also be increasing, which will help to recognize a clear decision boundary. Under 50 samples context, compared with the model without the DA module, the OA_o, AA_o, and K_o of the model with the DA module increased by 0.81%, 2.61%, and 1.12%, respectively. As for the 200 samples setting, the OA_o, AA_o, and K_o of the model with the DA module increased by 1.16%, 2.58%, and 1.50%, respectively.

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TABLE V
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR BOTH KNOWN AND UNKNOWN CLASSES ON WHU-HI-LONGKOU DATASET

Class No.	S=50					S=200				
	3DCNN	SSRN	MDL4OW	SSLR	SSEL	3DCNN	SSRN	MDL4OW	SSLR	SSEL
1	98.57	99.99	97.89	97.96	99.09	99.79	99.12	99.71	99.70	97.93
2	86.99	94.03	95.67	97.24	87.71	78.80	93.73	97.96	97.98	91.14
3	98.11	98.92	99.11	95.35	96.31	95.25	96.40	99.47	95.41	96.74
4	92.19	91.66	90.68	89.79	89.54	93.62	94.37	93.71	92.86	94.83
5	94.76	95.28	95.52	95.50	92.28	96.81	93.58	95.40	94.29	90.55
6	99.53	99.99	92.21	95.60	95.59	98.75	99.41	97.76	98.96	96.93
7	99.25	99.56	98.20	99.61	99.80	99.18	99.92	98.01	99.50	99.88
8	95.50	96.59	80.90	71.15	87.29	94.42	98.35	85.09	84.38	86.36
Novel	0.00	0.00	51.33	42.34	**76.44**	0.00	0.00	54.33	44.73	**81.68**
OA_c	96.17	**96.73**	94.64	94.71	95.04	96.38	**97.51**	96.43	96.62	96.70
AA_c	95.61	**97.00**	93.77	92.77	93.45	94.58	**96.86**	95.88	95.39	94.30
K_c	94.92	**95.67**	92.93	93.02	93.45	95.19	**96.68**	95.28	95.53	95.62
OA_o	93.68	94.23	93.53	93.37	**94.56**	93.89	94.98	95.35	95.29	**96.31**
AA_o	84.99	86.23	89.05	87.17	**91.56**	84.07	86.10	91.27	89.76	**92.89**
K_o	91.74	92.45	91.58	91.34	**92.93**	92.00	93.41	93.94	93.85	**95.18**

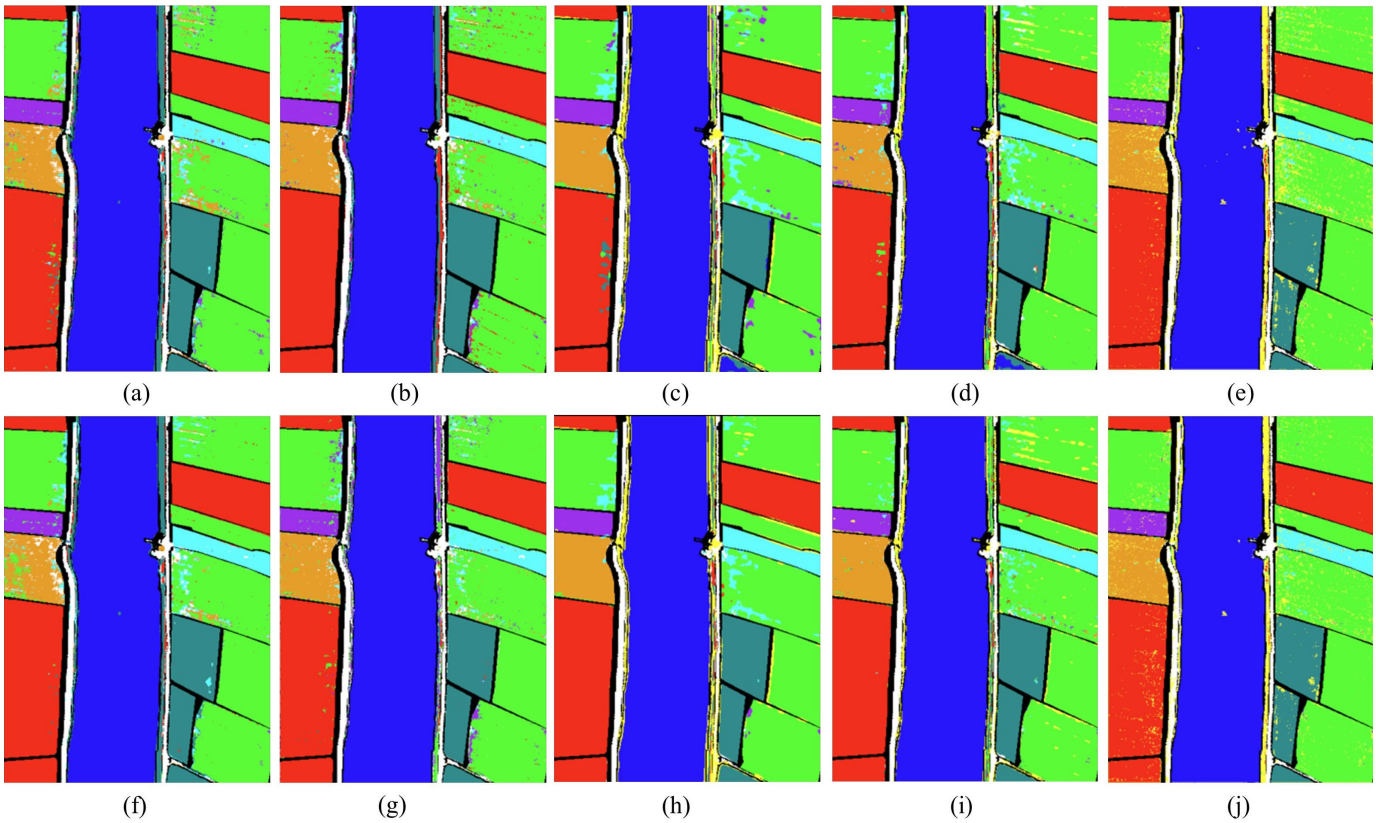


Fig. 9. Classification results of different methods for the WHU-Hi-LongKou dataset. (a) 3DCNN ($S = 50$). (b) SSRN ($S = 50$). (c) MDL4OW ($S = 50$). (d) SSLR ($S = 50$). (e) SSEL ($S = 50$). (f) 3DCNN ($S = 200$). (g) SSRN ($S = 200$). (h) MDL4OW ($S = 200$). (i) SSLR ($S = 200$). (j) SSEL ($S = 200$).

For the Salinas dataset, we also report the performance of the model with and without the DA strategy from the overall accuracy, average accuracy, and kappa. When the DA module was added, the accuracy of the model also increased in the open-set environment under different sample proportions. Specifically, compared with the model without the DA module, when $S = 50$, the OA_o, AA_o, and K_o. of the model with the DA strategy are increased by 0.53%, 0.70%, and 0.58%,

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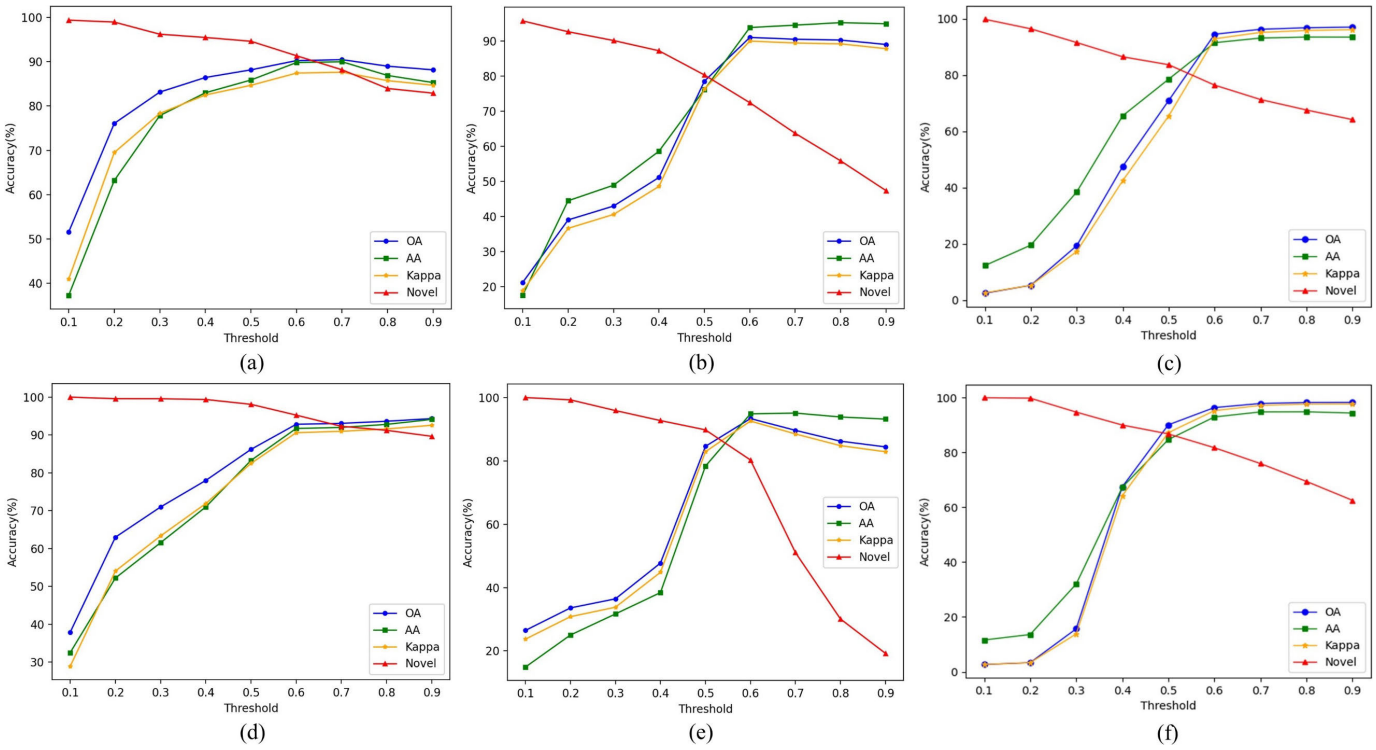


Fig. 10. Sensitivity analysis of the uncertainty threshold. (a) Pavia University ($S = 50$). (b) Salinas ($S = 50$). (c) WHU-Hi-LongKou ($S = 50$). (d) Pavia University($S = 200$). (e) Salinas ($S = 200$). (f) WHU-Hi-LongKou ($S = 200$).

TABLE VI
COMPARISON RESULTS OF THE GAN MODEL WITH DIFFERENT MODULES ON THREE DATASETS

Index	S=50		S=200	
	Without DA	With DA	Without DA	With DA
	Pavia University			
Novel	86.80	**91.34**	92.40	**95.25**
OA_o	89.43	**90.24**	91.66	**92.82**
AA_o	87.19	**89.80**	89.14	**91.72**
K_o	86.30	**87.42**	89.11	**90.61**
	Salinas			
Novel	71.82	**72.44**	76.34	**80.60**
OA_o	90.50	**91.03**	92.17	**93.37**
AA_o	93.15	**93.85**	92.36	**94.85**
K_o	89.41	**89.99**	91.25	**92.58**
	WHU-Hi-LongKou			
Novel	73.10	**76.44**	74.87	**81.68**
OA_o	92.65	**94.56**	95.85	**96.31**
AA_o	85.79	**91.56**	90.83	**92.89**
K_o	90.44	**92.93**	94.57	**95.18**

respectively. When $S = 200$, the OA_o, AA_o, and K_o. of the model with the DA strategy are increased by 1.20%, 2.49%, and 1.33%, respectively.

Similarly, the model with the DA module performed better in terms of four evaluation metrics (Novel, OA_o, AA_o, and K_o) on the WHU-Hi-LongKou dataset. Specifically, compared with the model without the DA module, when $S = 50$, the OA_o, AA_o, and K_o of the model with the DA strategy are increased by 1.91%, 5.77%, and 2.49%, respectively. When $S = 200$, the OA_o, AA_o, and K_o. of the model with the DA strategy are increased by 0.46%, 2.06%, and 0.61%, respectively.

C. Analysis of Patch Size in Our Proposed Method

In an open-set classification task of HSIs, the input size of the patch affects the spectral-spatial feature extraction ability of the model. To evaluate the effect of patch size on spectral-spatial feature extraction, we performed accuracy comparisons between different patch sizes of 1×1, 3×3, 5×5, and 7×7 on three datasets under different sample proportions, and the classification results on three datasets are shown in Tables VII–IX. In general, the patch-based open-set classification method of HSIs outperforms the pixel-based method, which is mainly because the relationship between adjacent pixels can provide richer context information. This helps the model to understand the structure and semantics of the image so that the patch-based model can classify the image more accurately. The model, moreover, achieves the best performance on three datasets under different sample proportions when the patch size is set to 5×5, and the accuracy decreases from that of the 5×5 patch for both closed-set and open-set environments when the patch size is 7×7, which may be explained by the larger patch size introducing additional information, which interferes with the open-set classification. As for the 3×3 setting, the model also performs worse than that of 5×5 under both closed-set and open-set environments, which may be due to the small patch

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TABLE VII
CLASSIFICATION RESULTS (%) ON THE UNIVERSITY OF PAVIA DATASET WITH DIFFERENT PATCH SIZE

	S=50				S=200			
Patch	1×1	3×3	5×5	7×7	1×1	3×3	**5×5**	7×7
Novel	83.95	88.38	**91.34**	87.54	86.58	90.59	**95.25**	93.45
OA_c	86.54	88.75	**90.21**	89.68	88.06	89.81	**92.77**	90.01
AA_c	81.76	87.51	**89.61**	87.58	87.80	89.72	**91.27**	88.80
K_c	82.32	85.25	**87.21**	86.43	84.50	86.61	**90.40**	86.91
OA_o	86.47	88.74	**90.24**	89.63	88.02	89.83	**92.82**	90.10
AA_o	82.00	87.61	**89.80**	87.57	87.65	89.80	**91.72**	89.31
K_o	82.49	85.44	**87.42**	86.55	84.67	86.81	**90.61**	87.19

TABLE VIII
CLASSIFICATION RESULTS (%) ON SALINAS DATASET WITH DIFFERENT PATCH SIZE

	S=50				S=200			
Patch	1×1	3×3	5×5	7×7	1×1	3×3	5×5	7×7
Novel	59.59	64.87	**72.44**	71.59	64.34	72.42	**80.60**	76.90
OA_c	92.20	94.91	**94.96**	93.79	91.46	94.51	**96.14**	93.23
AA_c	94.52	94.93	**95.38**	95.32	92.80	91.95	**95.89**	92.31
K_c	91.21	94.22	**94.28**	92.99	90.36	93.77	**95.61**	92.33
OA_o	86.51	89.67	**91.03**	89.92	86.73	90.66	**93.37**	90.38
AA_o	92.19	92.92	**93.85**	93.74	90.90	90.65	**94.85**	91.28
K_o	85.10	88.50	**89.99**	88.80	85.29	89.58	**92.58**	89.27

TABLE IX
CLASSIFICATION RESULTS (%) ON WHU-HI-LONGKOU DATASET WITH DIFFERENT PATCH SIZE

	S=50				S=200			
Patch	1×1	3×3	5×5	7×7	1×1	3×3	5×5	7×7
Novel	63.29	72.35	**76.44**	70.27	75.61	73.06	**81.68**	77.51
OA_c	92.39	93.97	**95.04**	92.76	90.91	95.38	**96.70**	95.38
AA_c	85.39	88.90	**93.45**	90.65	91.27	93.20	**94.30**	91.29
K_c	89.98	92.03	**93.45**	90.48	88.18	93.87	**95.62**	93.87
OA_o	91.63	93.41	**94.56**	92.18	90.51	94.80	**96.31**	94.92
AA_o	82.93	87.06	**91.56**	88.38	89.53	90.96	**92.89**	89.76
K_o	89.15	91.42	**92.93**	89.86	87.84	93.21	**95.18**	93.36

size not providing sufficient spatial information to distinguish different classes.

D. Sensitivity of the Uncertainty Threshold

The uncertainty threshold δ is an important factor that has an essential effect on the open-set classification results of HSIs and should be set carefully. We show the threshold analysis that includes four evaluation metrics (Novel, OA_o, AA_o, and K_o) under different sample proportions context on three datasets in Fig. 10. Specifically, we set the uncertainty threshold from 0.1 to 0.9 and examine the performance of the model on three datasets under different sample proportions. We can clearly see that the proposed SSEL method is somewhat sensitive to the uncertainty threshold. As the threshold increases, although the overall performance of the model becomes stronger, the ability of the model to recognize unknown classes gradually becomes weaker, which is mainly because large intraclass variation and interclass similarity are common in HSIs. Fortunately, when the uncertainty threshold $\delta \geq 0.6$, the overall accuracy of our proposed model becomes stable, and the ability of the model to recognize unknown classes can also achieve acceptable performance when $\delta = 0.6$; therefore, we finally set 0.6 as the uncertainty threshold.

TABLE X
TRAINING (PEER 100 EPOCHS) AND INFERENCE TIME (S)

Method	Pavia University		Salinas		WHU-Hi-LongKou	
	Train	Predict	Train	Predict	Train	Predict
3DCNN	46	3	109	9	88	187
SSRN	200+	200+	200+	200+	200+	200+
MDL4OW	48	46	49	37	69	81
SSLR	76	80	54	53	42	85
SSEL	45	14	51	12	67	29

E. Computational Cost

Finally, we report the training time (peer 100 epochs) and inference time of the proposed method and the comparison methods on three datasets under the 50 samples context in Table X. As can be seen from this table, the proposed SSEL method achieves competitive performance in both training and inference phases, especially in the inference phase. This is mainly because only the discriminator is used in the inference phase; furthermore, due to the spectral residual blocks in the SSRN, it tends to require a larger amount of computational power than other methods.

V. CONCLUSION

In this article, a novel SSEL network is proposed for open-set HSI classification, which we believe is the first study to introduce evidential theory into open-set HSI classification. Conventional HSI classification assumes that the class distribution in the training and test stages is consistent and there will be no unknown classes in the data. This assumption is risky since the Earth's surface is diverse and complex, and it is hard to include all the potential classes in the training set. To tackle this issue, a novel multiclass spectral-spatial GAN that incorporates evidential theory is proposed for HSI classification in the open world, where the generative model GAN is used to expand samples and extract spectral-spatial features for multiclass evidence acquisition, while the collected evidence will be used to closed-set classification and uncertainty estimation of classification results. Finally, the samples with high uncertainty will be rejected as an unknown class. Extensive experiments show that the proposed method significantly improves the performance of unknown

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class recognition while maintaining satisfactory classification accuracy for known classes. Comprehensive comparison experiments also demonstrate that the performance of the proposed method is better than existing state-of-the-art closed-set and open-set classification methods of HSIs. In the future, few-shot HSI classification under open-set environments will be explored to further reduce the reliance on labeled data.

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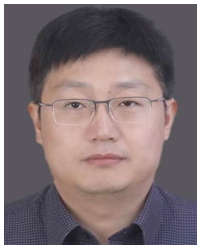
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